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A MGT Representation Analysis

A.1 Representation Quality Metrics

We briefly introduce the representation-quality metrics used in our analysis and provide their formulae below. Following [Skean et al. \(2025\)](#), we group these metrics into *information-theoretic*, *geometric*, and *invariance-based* categories. Our analysis focuses on the first two categories, which characterize the complexity, dimensionality, and geometry of MGT and HWT representations.

1. **Entropy** ([Sanchez Giraldo et al., 2015](#)) measures how much information a representation contains. Higher values indicate more diverse and information-rich features, whereas lower values suggest more redundant and compressed representations. For activations at layer l , let $Z^{(l)} \in \mathbb{R}^{d \times N}$ denote the representation matrix and define the corresponding Gram matrix as:

$$K^{(l)} = Z^{(l)} Z^{(l)\top}.$$

Let $\{\lambda_i(K^{(l)})\}$ denote the non-negative eigenvalues of $K^{(l)}$. For any order $\alpha > 0$, the entropy is defined as:

$$\text{Entropy}(Z^{(l)}) = \frac{1}{1 - \alpha} \log \sum_{i=1}^r \left(\frac{\lambda_i(K^{(l)})}{\text{tr}(K^{(l)})} \right)^\alpha,$$

where $r = \text{rank}(K^{(l)}) \leq \min(N, d)$. Following [Skean et al. \(2025\)](#), we compute the von Neumann entropy by setting $\alpha = 1$.

2. **Effective Rank** ([Roy and Vetterli, 2007](#); [Wei et al., 2024](#)) measures the dimensional complexity of representations. Higher values indicate more distributed and noisy feature representations, while lower values suggest stronger compression and more compact representations. For activations at layer l , represented by the matrix $A^{(l)} \in \mathbb{R}^{d \times N}$ with N samples and hidden dimension d , the effective rank is defined as:

$$\text{eR}(A^{(l)}) = \exp \left(- \sum_{i=1}^Q \frac{\sigma_i}{\sum_{j=1}^Q \sigma_j} \log \frac{\sigma_i}{\sum_{j=1}^Q \sigma_j} \right),$$

where $Q = \min(N, d)$ and $\sigma_1, \dots, \sigma_Q$ denote the singular values of $A^{(l)}$.

3. **Anisotropy** ([Razzhigaev et al., 2024](#)) measures the degree to which activations concentrate along specific directions in latent space. Higher anisotropy indicates that activations are strongly aligned with a smaller set of dominant directions, whereas lower values indicate more uniformly distributed representations. To compute anisotropy, let $X^{(l)} \in \mathbb{R}^{N \times d}$ denote the centered activations at layer l , where $\sigma_1, \dots, \sigma_k$ are the singular values of $X^{(l)}$. The anisotropy score is defined as:

$$\text{Anisotropy}(X^{(l)}) = \frac{\sigma_1^2}{\sum_{i=1}^k \sigma_i^2},$$

where $k = \min(N, d)$.

4. **Intrinsic Dimension** ([Facco et al., 2017](#)) estimates the minimum number of dimensions required to describe the local structure of representations without substantial information loss. Higher values indicate richer and more complex latent structures, whereas lower values suggest that representations lie on simpler and lower-dimensional manifolds. We estimate intrinsic dimensionality using the Two-Nearest Neighbor estimator proposed by [Facco et al. \(2017\)](#), implemented via the `scikit-dimension` package.⁸ We refer readers to the original work for details on the estimator and its formula.

A.2 Extended Analysis

In Figure 1, we visualized the activations of human- and machine-written Wikipedia text projected onto the first two principal components. Figure 7 extends this analysis to the Reddit, PeerRead, and ArXiv subsets of M4GT ([Wang et al., 2024](#)). Across all domains, we observe the same qualitative pattern: human- and machine-written texts occupy distinct regions of latent space and remain linearly separable throughout the middle and later layers.

In Figure 8, we visualize activations of human- and machine-written text from different domains (a) and generators (b) in M4GT ([Wang et al., 2024](#)) using t-SNE. In both plots, each domain or generator occupies a distinct local region of latent space, which further separates into human- and machine-written clusters. Although t-SNE is nonlinear and

⁸<https://github.com/scikit-learn-contrib/scikit-dimension>